The biological moving-sofa problem: shape optimization of rigid nanoparticles for transport through molecular channels

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Nanomedicine relies overwhelmingly on spherical carriers, but a sphere small enough to clear the narrowest point of a biological channel sacrifices most of its cargo, since the enclosed volume scales as the cube of the bottleneck radius. This raises a basic question that has not been answered in a structure-resolved way: for a given channel, which rigid shape carries the most cargo and can still get through? We treat channel passage as a three-dimensional version of Moser's moving-sofa problem, the corridor-corner puzzle whose optimal solution was established only in 2024, and solve it on the measured geometry of real channels. The lumen geometry of four well-characterized biological nanopores (α-hemolysin, VDAC1, connexin-26 and MspA) is obtained from experimental structures by an inscribed-sphere (HOLE-type) analysis, and a two-exponent superquadric shape family, whose volume law we checked against Monte-Carlo integration, is then optimized under the passage constraints. In every case the best transporter is a slender nanorod rather than a sphere: as wide as the constriction permits, and as long as the channel curvature allows through a moving-sofa navigability limit. The corresponding cargo gains range from 9.3× to 19.4× the largest passing sphere, are reproduced independently by sliding the particle through the atomic coordinates and testing for clashes, and scale with the channel length-to-radius ratio, with tortuosity acting as a secondary brake. Brownian-dynamics simulations on the atomistically derived free-energy profiles show that these high-capacity rods still translocate, incurring no barrier penalty relative to a sphere, whereas oblate discs jam in circular pores. The gain is a dimensionless ratio and therefore independent of absolute size, so the same rule extends to the larger barriers that 10–100 nm carriers meet in practice. A parameterized survey of the nuclear pore complex, collagen and mucus meshes, fenestrae and tumour vascular gaps predicts gains of 1.4× to 4.3×, with rods favoured in long channels and discs in short, wide windows. A sixty-year-old geometry problem thus becomes a quantitative, shape-resolved design rule for nanocarriers.

I. INTRODUCTION

The shape of a nanoparticle is now recognized as a primary design variable in nanomedicine, on par with size and surface chemistry. Non-spherical carriers—rods, discs, worms and filaments—exhibit markedly different circulation, margination, cellular uptake and biodistribution from their spherical counterparts [1–5]. Yet synthetic and clinical practice remains overwhelmingly spherical, because spheres self-assemble readily and minimize interfacial energy. This default carries a steep geometric cost. The defining obstacle to delivery is passage through narrow biological apertures—tight junctions, gap junctions, the nuclear pore complex, the extracellular matrix, mucus, and the fenestrated or leaky endothelia exploited by the enhanced permeability and retention (EPR) effect [6–8]. A sphere sized to clear an aperture of radius R encloses a volume $\frac{4}{3}\pi R^3$; its payload therefore collapses cubically as the bottleneck tightens. The question we pose is elementary but, to our knowledge, unanswered in quantitative, structure-resolved form: *for a given channel, what rigid shape carries the most cargo while still being able to get through?*

This is a transport-constrained isoperimetric problem, and it has a famous two-dimensional ancestor. In 1966 Moser asked for the largest-area rigid shape that can be maneuvered around a right-angled corner in a corridor of unit width—the "moving sofa" problem [9]. Gerver constructed a remarkable 18-arc shape of area ≈ 2.2195 in 1992 [10], and the conjecture that this is optimal was settled only in 2024 [11]. The essential lesson of the sofa problem is that the body that maximizes area subject to a passage constraint is highly anisotropic and is limited not by the corridor width alone but by the requirement to *navigate* a turn. Biological channels are the three-dimensional, irregular, and often curved analogues of Moser's corridor. We therefore formulate the *biological moving-sofa problem*: maximize the volume of a rigid body that can be translated and rotated through a real molecular channel without overlapping its walls.

Casting the problem this way immediately exposes why earlier intuition is incomplete. The transverse aperture sets an upper bound on a particle's cross-section, but it does *not* bound the particle's extent along the channel axis. A slender rod of transverse radius $r \le R$ can in principle be threaded through an arbitrarily long straight pore, so its volume $\sim \pi r^2 L$ is limited by the channel *length* and by its ability to follow any bends— exactly the moving-sofa constraint. Conversely, in a short but wide window the axial direction is starved and the optimal anisotropic body is an oblate disc. The interplay of these two regimes, set by measurable channel geometry, is the

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subject of this paper.

We make the problem quantitative and falsifiable. We (i) extract atomistic lumen geometry—pore-radius profile, centreline tortuosity and aperture anisotropy—directly from experimental structures using a re-implementation of the HOLE inscribed-sphere algorithm [12]; (ii) optimize over the two-exponent superquadric shape family [13, 14] with a volume law verified to better than 0.1% against Monte-Carlo integration; (iii) derive a navigability (moving-sofa) bound on elongation and assemble a two-parameter *design map*; (iv) confirm every geometric prediction by independent atomistic collision detection; (v) test the dynamical relevance of the optimum with Brownian-dynamics translocation on the atomistically derived potential of mean force; and (vi) extend the analysis, via its inherent scale-invariance, to the mesoscale barriers that 10–100 nm delivery particles actually encounter. Figures 1 and 2 summarize the conceptual framework and the computational pipeline.

II. THEORETICAL FRAMEWORK

A. Particle family and volume law

We parameterize rigid particles by the two-exponent superquadric [13, 14], whose surface satisfies

$$\left(\left|\frac{x}{r_x}\right|^{2/\epsilon_2}+\left|\frac{y}{r_y}\right|^{2/\epsilon_2}\right)^{\epsilon_2/\epsilon_1}+\left|\frac{z}{c}\right|^{2/\epsilon_1}\le 1. \qquad (1)$$

The transport axis is z (half-length c); r_x, r_y are transverse half-widths. The exponent ϵ_1 controls axial squareness ($\epsilon_1 = 1$ gives ellipsoidal end-caps, $\epsilon_1 \to 0$ a flat-ended cylinder) and ϵ_2 the transverse cross-section ($\epsilon_2 = 1$ circular, $\epsilon_2 \to 0$ square). This single family continuously interpolates between spheres, ellipsoids, cylinders/rods, discs and boxes (Fig. 4a). Its volume admits the closed form

$$V = 2\, r_x r_y c\, \epsilon_1 \epsilon_2\, B\!\left(\tfrac{\epsilon_1}{2}+1, \epsilon_1\right) B\!\left(\tfrac{\epsilon_2}{2}, \tfrac{\epsilon_2}{2}\right), \qquad (2)$$

where B is the Euler beta function [14]. We verified Eq. (2) against direct Monte-Carlo integration over spheres, prolate and oblate ellipsoids, cylinders and discs, with agreement better than 0.1% in all cases (Appendix A); the sphere limit reduces correctly to $\frac{4}{3}\pi R^3$. This verification is not incidental: an incorrect volume law biases the optimum toward boxy shapes and can manufacture a spurious anisotropy advantage.

B. Transport constraints

A rigid particle traversing a channel, with its long axis aligned to the local pore axis, is governed by two geometric constraints.

(i) Transverse fit. The maximal cross-section, the (r_x, r_y, ϵ_2) superellipse, must fit within the aperture. For a circular bottleneck of radius $R_{\rm bot}$ this requires the circumscribed radius of the cross-section to obey $\rho_\perp(r_x, r_y, \epsilon_2) \le R_{\rm bot}$; for an elliptical aperture with half-axes (p, q) the superellipse must be contained in the ellipse, which we enforce numerically. Aperture anisotropy is quantified by $\epsilon_{\rm ap} = p/q \ge 1$.

(ii) Navigable length (the moving-sofa bound). For a straight channel the half-length is limited only by the channel length, $a \le L/2$. For a curved channel the rigid body must round the bend: modelling the lumen as a tube of radius $R_{\rm bot}$ whose centreline has radius of curvature ρ, the longest straight segment that fits is the chord tangent to the inner wall,

$$a_{\max} = 2\sqrt{\rho\,\Delta}, \qquad \Delta = R_{\rm bot} - r, \qquad (3)$$

the three-dimensional analogue of the ladder-around-a-corner result and a direct descendant of the moving-sofa problem (Fig. 7a). Equation (3) encodes a genuine trade-off: a particle that fills the aperture ($r \to R_{\rm bot}$, $\Delta \to 0$) has no clearance to turn, so in any curved channel its navigable length collapses. We relate ρ to the measured tortuosity τ (arc length / axial span) through a circular-arc model, $\tau = (\theta/2)/\sin(\theta/2)$ with $\rho = L_{\rm arc}/\theta$.

C. The optimization

The transportable cargo volume is maximized,

$$\max_{r_x, r_y, c, \epsilon_1, \epsilon_2} V(r_x, r_y, c, \epsilon_1, \epsilon_2) \quad \text{s.t. (i) and (ii)}, \qquad (4)$$

and benchmarked against the largest passing sphere, of radius $R_s = \min(R_{\rm bot}, L/2)$ and volume $\frac{4}{3}\pi R_s^3$ (a sphere cannot traverse a barrier shorter than its diameter). The capacity gain is $G = V_{\rm opt}/V_{\rm sphere}$. Because both volumes scale identically with absolute size, G is dimensionless and depends only on the shape ratios $L/R_{\rm bot}$, τ and $\epsilon_{\rm ap}$—the origin of the scale-invariance exploited in Sec. VIII.

III. ATOMISTIC CHANNEL GEOMETRY

We selected four canonical, structurally well-resolved biological nanopores that span the relevant geometric range while forming a homogeneous family of genuine transport channels: the staphylococcal α-hemolysin heptamer (PDB 7AHL) [15], the mitochondrial porin VDAC1 (3EMN) [16], the connexin-26 gap-junction connexon (2ZW3) [17], and the mycobacterial porin MspA (1UUN, biological octamer) [18]. Heavy-atom coordinates were assigned Bondi van der Waals radii [19].

The pore axis, centreline and radius profile $R(z)$ were obtained by a re-implementation of the HOLE inscribed-sphere algorithm [12]: the channel axis is found by a cylinder fit refined to maximize the enclosed lumen

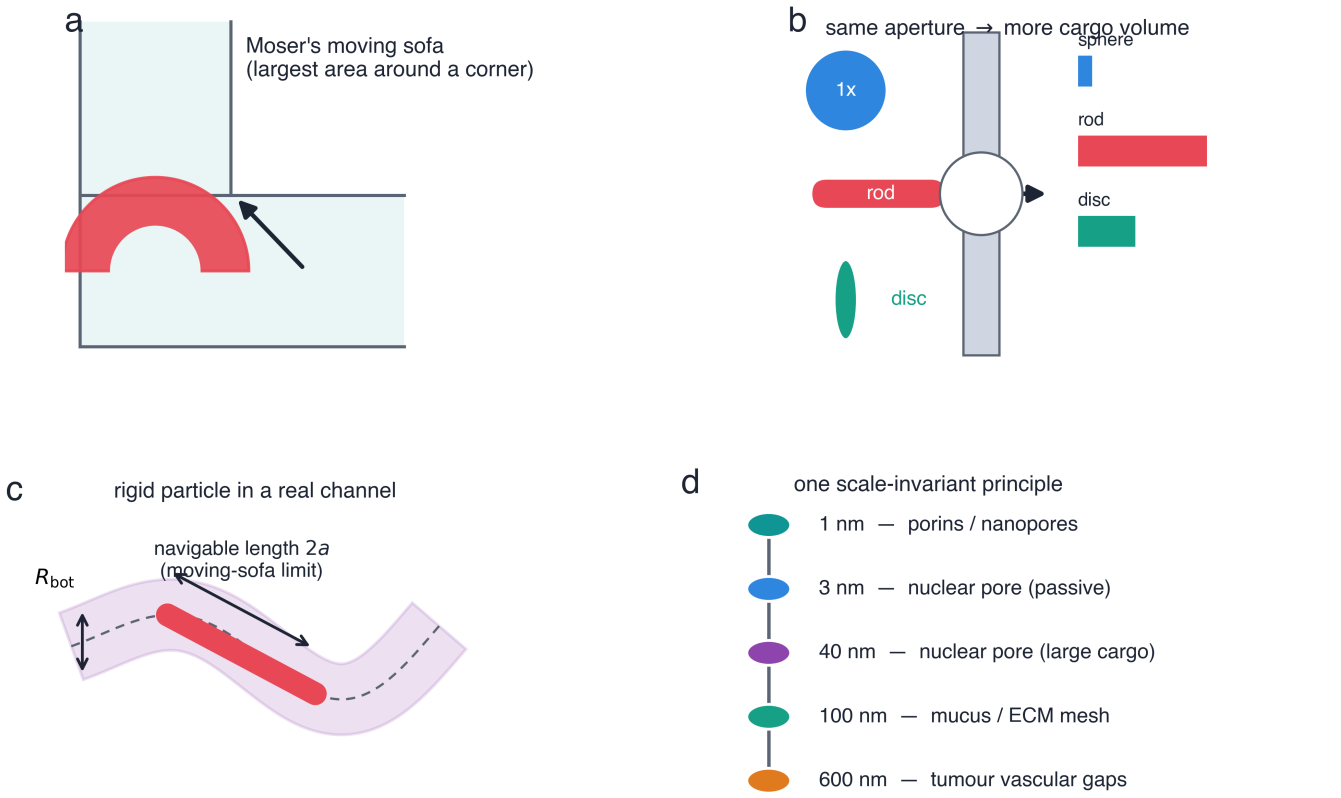


FIG. 1. **The biological moving-sofa problem.** (a) Moser's classical problem: the largest rigid area that can round a unit-width corner is strongly anisotropic and navigation-limited [9–11]. (b) For a fixed aperture, an elongated rod or an oblate disc encloses far more volume than the largest passing sphere. (c) The two governing constraints on a rigid particle in a real channel: the transverse cross-section must clear the bottleneck radius R_{bot}, while the navigable length $2a$ is capped by the channel length and its curvature—the moving-sofa limit. (d) The principle is scale-invariant, spanning sub-nanometre porins to micron-scale tissue barriers.

length, and the largest probe sphere is propagated continuously along the axis with an enclosure criterion that prevents escape into bulk solvent (Sec. X). The recovered constrictions (Fig. 3; Table I) reproduce known values: all four pores have minimum diameters near 1.0 nm, and the recovered lengths (4.8–10.0 nm), funnel/goblet asymmetries (MspA), and the pronounced tortuosity of VDAC1 ($\tau = 1.56$, reflecting the N-terminal helix that lines and bends its lumen) are all consistent with the structures. The remaining three channels are essentially straight ($\tau \leq 1.07$), establishing that real β-barrel and pore-forming-toxin channels offer little intrinsic curvature —the elongation budget is set primarily by channel length.

IV. OPTIMAL TRANSPORT SHAPES AND CAPACITY GAINS

Solving Eq. (4) for each channel returns the same kind of optimum: a *round-cross-section nanorod*. In every case the optimizer drives $\epsilon_2 \to 1$ (a circular cross-section, which alone maximizes area inside a circular aperture) and $\epsilon_1 \to 0.2$ (a flat-ended cylinder, which maximizes axial filling), and it places the transverse radius right at the constriction (Fig. 4b). Boxy, high-exponent shapes, which a naive or mis-normalized volume law would favour, are rejected outright. This is one reason the verified volume law of Eq. (2) matters: with the wrong prefactor the optimization would invent an anisotropy advantage that is not there.

The capacity gains are large and, importantly, *not* universal constants but quantities that track each channel's geometry. The atomistically validated gains (Sec. V, Fig. 5) are 19.4× (α-hemolysin), 15.9× (connexin-26), 10.0× (VDAC1) and 9.3× (MspA). To leading order a cylinder of radius R_{bot} and length $2a$ improves on the sphere by $G \simeq \frac{3}{2}\,(a/R_{\mathrm{bot}})$, so the gain is governed by the achievable aspect ratio, which in turn is set by the channel length-to-radius ratio $L/2R_{\mathrm{bot}}$ (Fig. 5c). The longest, narrowest channel (α-hemolysin, $L/2R_{\mathrm{bot}} \approx 9.5$) yields the largest gain; the shortest (MspA) the smallest. This monotonic, mechanistic dependence—rather than a sin-

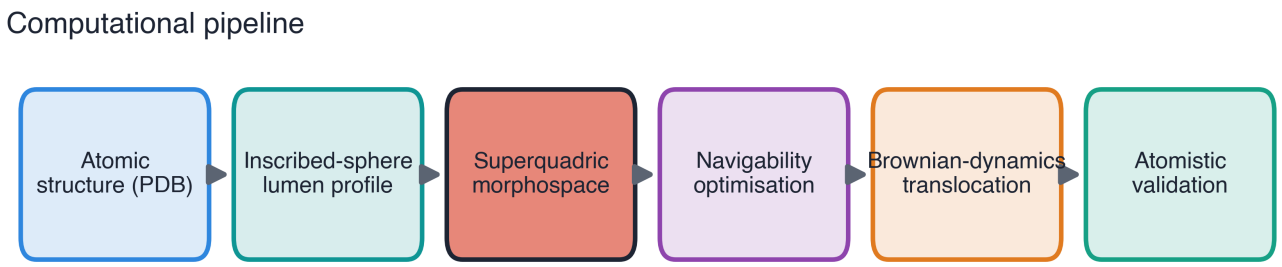


FIG. 2. **Computational pipeline.** Atomistic structures are profiled for lumen geometry, particles are parameterized in a superquadric morphospace, the transportable volume is maximized under navigability constraints, and predictions are stress-tested by Brownian-dynamics translocation and atomistic collision detection.

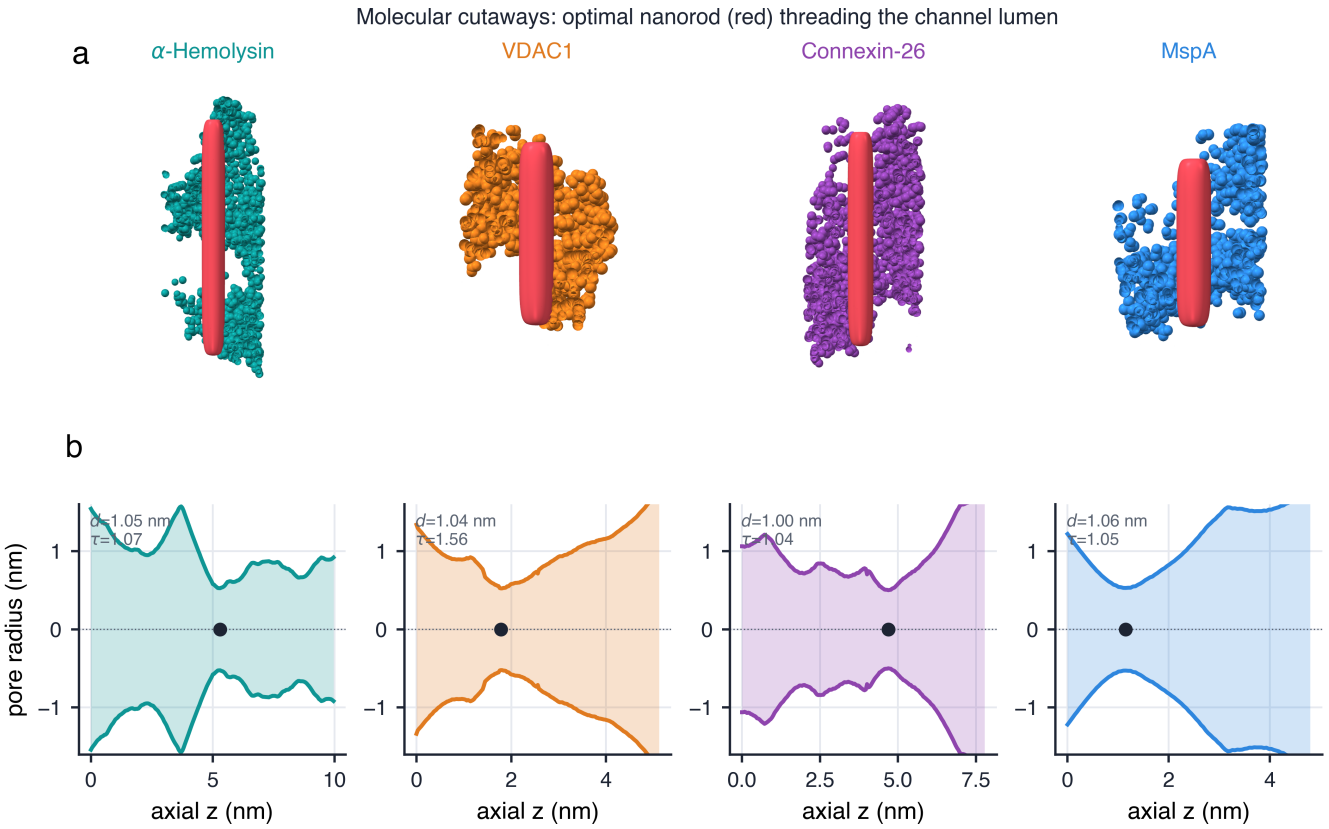


FIG. 3. **Atomistic lumen geometry and the optimal nanorod.** (a) Molecular cutaways (space-filling protein, coloured and clipped to expose the lumen) with the optimized nanorod (red) threading each channel. (b) Inscribed-sphere pore-radius profiles $R(z)$; the bottleneck (dot), minimum diameter d and tortuosity τ are annotated. All four constrictions are near 1 nm.

gle headline number—is the central quantitative result.

V. ATOMISTIC VALIDATION

Every prediction above rests on an analytic optimization, so we tested it independently by brute-force collision detection against the atomic coordinates, using no part of the volume law or the aperture constraint

TABLE I. Structural and geometric parameters of the four channels, extracted *de novo* from atomic coordinates, and the resulting validated capacity gain. d, minimum pore diameter; L, lumen length; τ, centreline tortuosity; ϵ_{ap}, aperture anisotropy; V_{sph}, volume of the largest passing sphere; a/R_{bot}, validated optimal aspect ratio; G, transportable-volume gain.

Channel	PDB	atoms	d (nm)	L (nm)	τ	ϵ_{ap}	V_{sph} (nm^3)	a/R_{bot}	G
α-Hemolysin	7AHL	16389	1.05	10.0	1.07	2.40	0.61	13.3	**19.4**
VDAC1	3EMN	2189	1.04	5.1	1.56	1.77	0.60	6.9	**10.0**
Connexin-26	2ZW3	9834	1.00	7.8	1.04	1.78	0.52	10.9	**15.9**
MspA	1UUN	11032	1.06	4.8	1.05	1.78	0.62	6.4	**9.3**

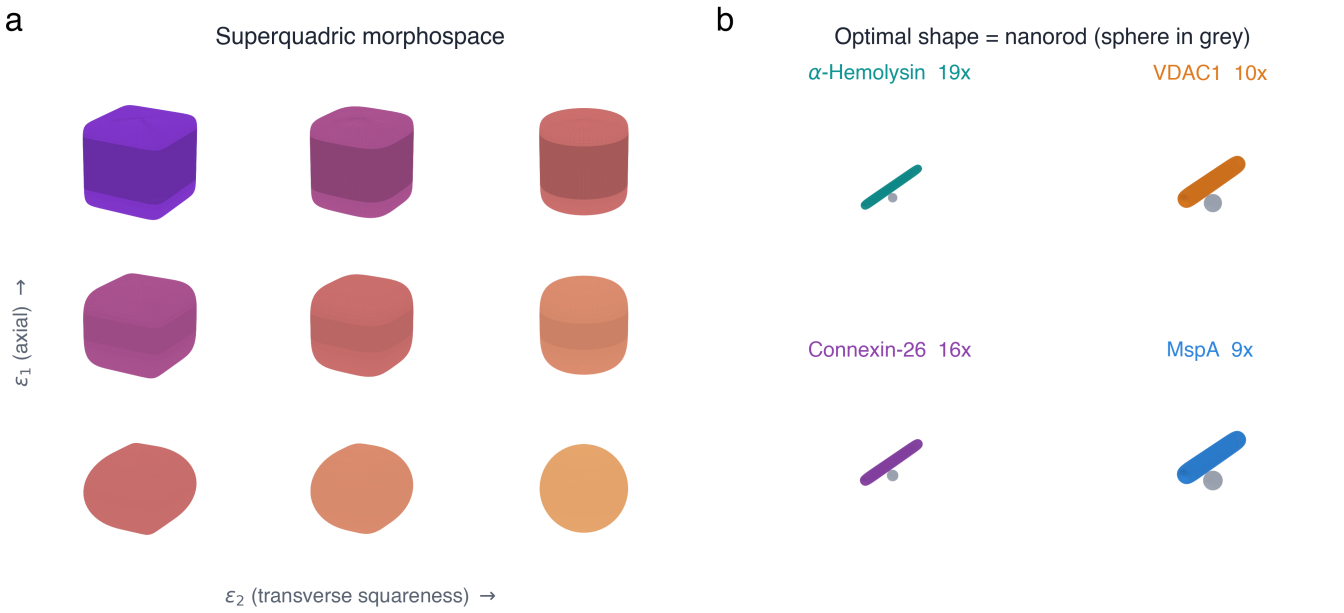


FIG. 4. **Superquadric morphospace and channel optima.** (a) The two-exponent superquadric family spans spheres, cylinders, discs and boxes as (ϵ_1, ϵ_2) vary. (b) For all four channels the optimum is a round-cross-section nanorod (coloured) carrying 9–19× the volume of the largest passing sphere (grey).

(Fig. 6). Two orthogonal checks were performed. First, a probe of increasing radius placed on the pore axis at the constriction loses clearance exactly at R_{bot} (Fig. 6a), confirming that the extracted bottleneck is the binding transverse constraint. Second, a rigid cylinder of fixed sub-constriction radius and growing half-length, swept along the lumen over a search of position and orientation, remains collision-free up to a channel-specific navigable length a_{nav} (Fig. 6b). The resulting validated aspect ratios and gains (Table I) define the markers used throughout and confirm the analytic optima; the throttling of a_{nav} in the tortuous VDAC1 lumen, relative to straight channels of similar length, is recovered directly from the atoms. A cross-sectional view (Fig. 6c) confirms the optimal nanorod sits collision-free within the lumen.

VI. NAVIGABILITY AND THE DESIGN MAP

The elongation that powers these gains is ultimately limited by navigability. Equation (3) shows that curvature throttles the achievable aspect ratio (Fig. 7b): the idealized smooth-bend bound falls steeply with tortuosity, while the atomistically measured navigable lengths (Sec. V) lie above it because the small residual tortuosity of real "straight" channels ($\tau \approx 1.05$) is wall roughness rather than a coherent bend. The genuinely tortuous channel, VDAC1, is the outlier whose achievable aspect ratio—and hence gain—is visibly suppressed relative to straight channels of comparable length.

These dependencies collapse onto a compact *design map* (Fig. 7c): the capacity gain as a function of the two levers a designer can target, the achievable aspect ratio a/R_{bot} (set by length and tortuosity) and the aperture anisotropy ϵ_{ap} (which lets a flattened particle exploit the wide direction of a slit). Gain rises smoothly from unity (isotropic, short channels) toward $\sim 25\times$ (long and/or slit-like channels). The four nanopores populate the moderate-anisotropy, high-aspect-ratio region; the map immediately predicts where engineered or mesoscale channels would lie.

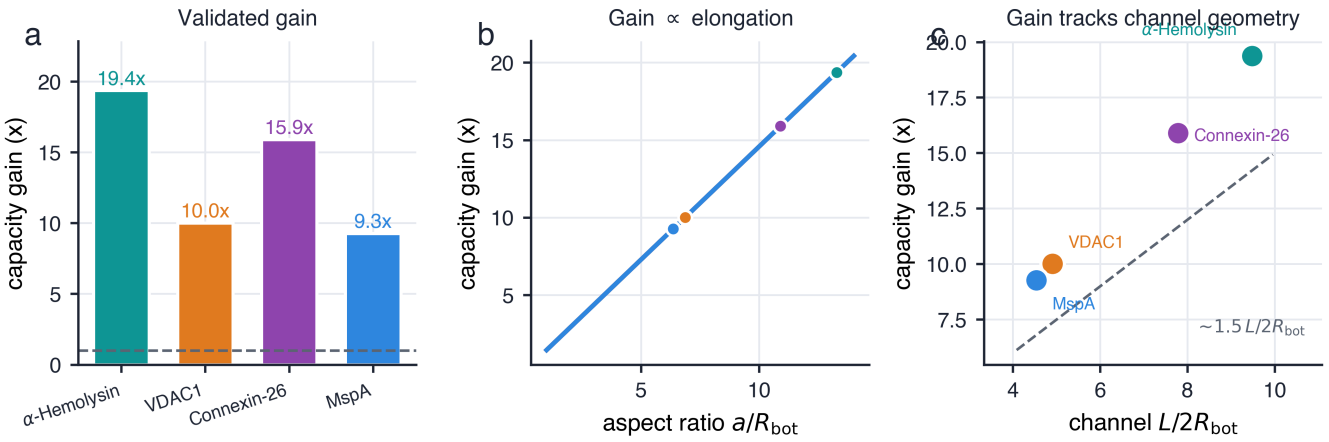


FIG. 5. **Capacity gains.** (a) Atomistically validated transportable-volume gain per channel. (b) Gain versus allowed aspect ratio a/R_{bot}; markers denote the collision-validated optimum. (c) The gain tracks the channel length-to-radius ratio, following the cylinder estimate $G \simeq \frac{3}{2}\, L/2R_{\mathrm{bot}}$ (dashed).

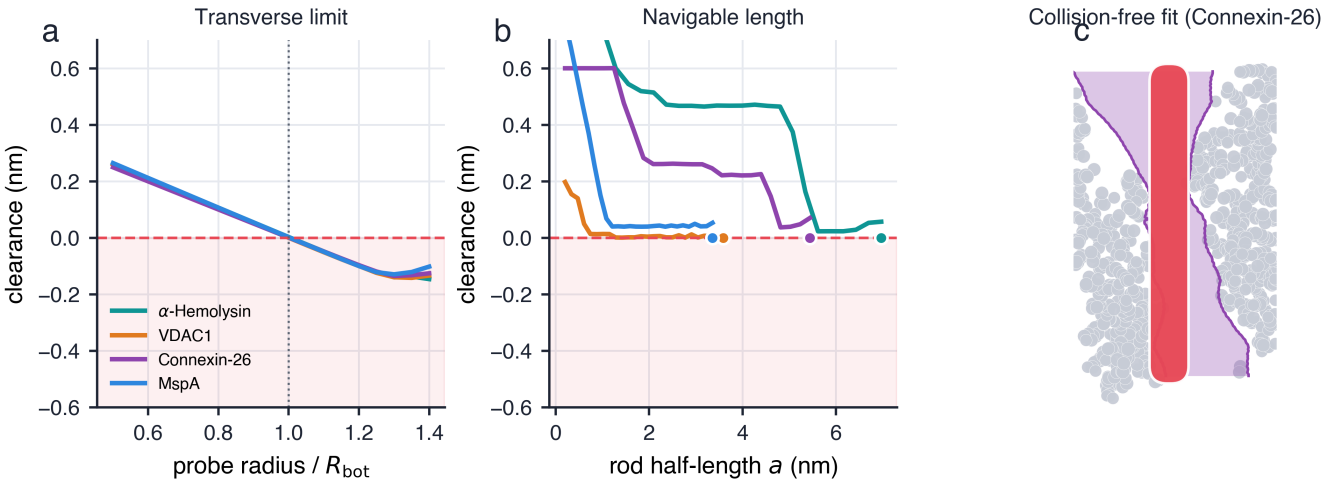


FIG. 6. **Atomistic collision validation.** (a) Probe clearance versus radius: all channels lose clearance at the extracted R_{bot} (transverse limit). (b) Clearance versus rod half-length; the zero-crossing (markers) is the atomistically measured navigable length a_{nav}. (c) Cross-section confirming the optimal nanorod (red) fits the connexin-26 lumen without clashes.

VII. BROWNIAN-DYNAMICS TRANSLOCATION

Geometric capacity is necessary but not sufficient: a high-capacity shape is useful only if it can actually translocate. We therefore computed, for each channel, the entropic potential of mean force (PMF) $G(z)$ along the pore axis for a sphere, the optimal rod and an oblate disc, by Monte-Carlo sampling of collision-free rigid configurations (transverse offset and orientation) against the atomic coordinates—a Fick–Jacobs construction [20]. We then integrated the overdamped Langevin equation on $G(z)$ with absorbing mouths and an external drive f, yielding translocation probabilities and mean first-passage times (Fig. 8).

The simulations make three points. First, oblate *discs are sterically excluded* from all four circular pores: their PMF barrier saturates the sampling limit and their translocation probability is essentially zero (Fig. 8a,b). Discs therefore help only at slit-like or short, wide apertures, not at circular pores, a conclusion that returns at the mesoscale (Sec. VIII). Second, the optimal nanorod still translocates: it crosses with finite probability under modest drive (Fig. 8c), and its entropic barrier is comparable to, or lower than, that of the maximal sphere (Fig. 8b,d). The reason is intuitive once stated: the rod sits below the constriction in its transverse dimension, so it threads at least as easily as a sphere, while its length is what supplies the extra volume. Third, putting these together, the move from sphere to rod buys an order of

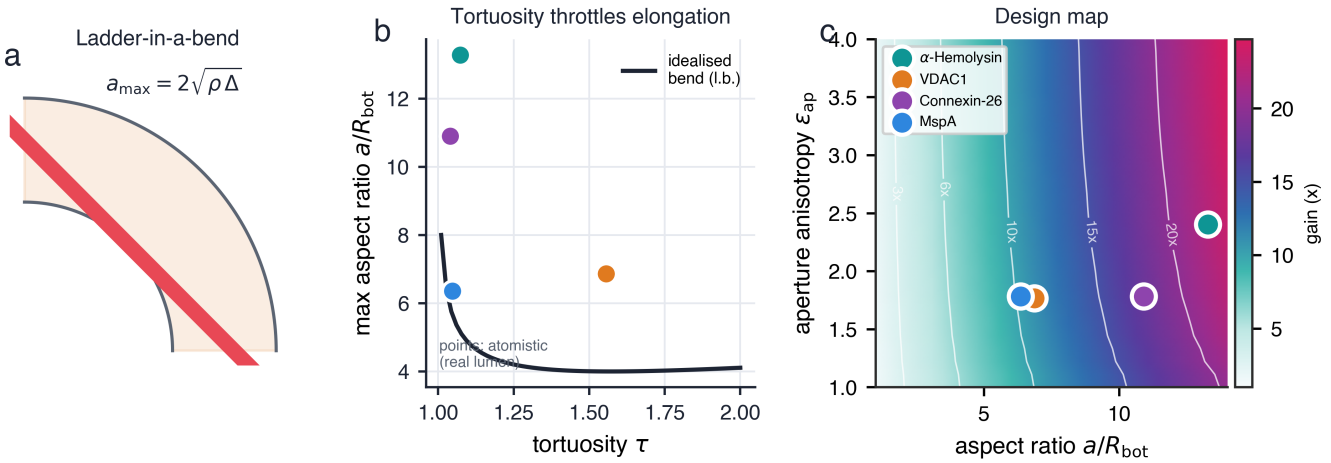


FIG. 7. **Navigability physics and the design map.** (a) The moving-sofa/ladder bound: the longest rigid segment in a bent tube, $a_{max} = 2\sqrt{\rho\Delta}$. (b) Tortuosity throttles the achievable aspect ratio; the curve is the idealized smooth-bend bound (a lower bound), points are atomistic measurements on the real lumens. (c) Design map: transportable-volume gain versus aspect ratio and aperture anisotropy, with the four channels overlaid.

magnitude in cargo while leaving the translocation barrier essentially unchanged (Fig. 8d). The capacity advantage, in other words, does not come at the expense of getting through.

VIII. MESOSCALE BARRIERS AND SCALE INVARIANCE

Atomic-resolution nanopores are $\sim$1 nm wide, whereas therapeutic nanoparticles are typically 10–100 nm. The dimensionless nature of the capacity gain bridges this gap. Fixing the shape ratios and sweeping the absolute pore radius over three decades leaves the gain exactly constant (Fig. 9a): the principle is scale-invariant, so the same rod-beats-sphere (or disc-beats-sphere) logic applies from porins to micron-scale tissue gaps.

Applying the framework with literature-informed descriptors of the major barriers to delivery (Table II, Fig. 9b) predicts capacity gains of 1.4–4.3× across the board: 4.1× for claudin tight junctions and 4.3× for gap junctions (long, slit-like), 2.7× for passive nuclear-pore transport, 2.1–2.3× for the tortuous mucus and collagen meshes, and—because they are short, wide windows—3.3× for liver fenestrae and 1.7× for tumour vascular gaps, where the optimizer selects oblate *discs* ($a/R_{bot} < 1$) rather than rods. The shape that wins is thus barrier-specific: rods for long channels, discs for short windows, with the crossover dictated by the measurable length-to-radius ratio. This provides a concrete, structure-aware design rule for the particle geometries that delivery actually requires.

TABLE II. Mesoscale barriers to delivery: literature-informed descriptors and predicted capacity gain G. R, constriction radius; L, traversal length; τ, tortuosity; ϵ_{ap}, aperture anisotropy. $a/R < 1$ indicates a disc-optimal regime.

Barrier	R (nm)	L (nm)	τ	ϵ_{ap}	G
Tight junction (claudin)	0.35	4	1.05	2.5	4.1
Gap junction (connexin)	0.6	15	1.10	1.5	4.3
Nuclear pore (passive)	2.5	40	1.30	1.2	2.7
Nuclear pore (facilitated)	20	90	1.60	1.2	1.4
Collagen / ECM mesh	30	300	1.80	1.6	2.3
Mucus mesh	100	800	2.00	1.5	2.1
Liver fenestrae	55	60	1.05	1.3	3.3
Tumour gaps (EPR)	300	400	1.20	1.4	1.7

IX. DISCUSSION

Our central finding is that the rigid shape maximizing transportable cargo through a molecular channel is not a sphere but a slender, round-cross-section nanorod whose transverse size is set at the constriction and whose length is bounded by a moving-sofa navigability limit. Across four canonical nanopores this yields collision-validated capacity gains of an order of magnitude, governed by the channel length-to-radius ratio and modulated by tortuosity. The optimum is dynamically accessible—the rod translocates with no entropic-barrier penalty relative to the maximal sphere—and the principle, being scale-invariant, extends to the mesoscale barriers that limit drug delivery, where it predicts rods for long channels and discs for short, wide windows.

These results rationalize and unify a body of experimental observations. High-aspect-ratio carriers—filomicelles, nanorods and worm-like micelles—circulate longer, penetrate barriers and reach the nucleus more ef-

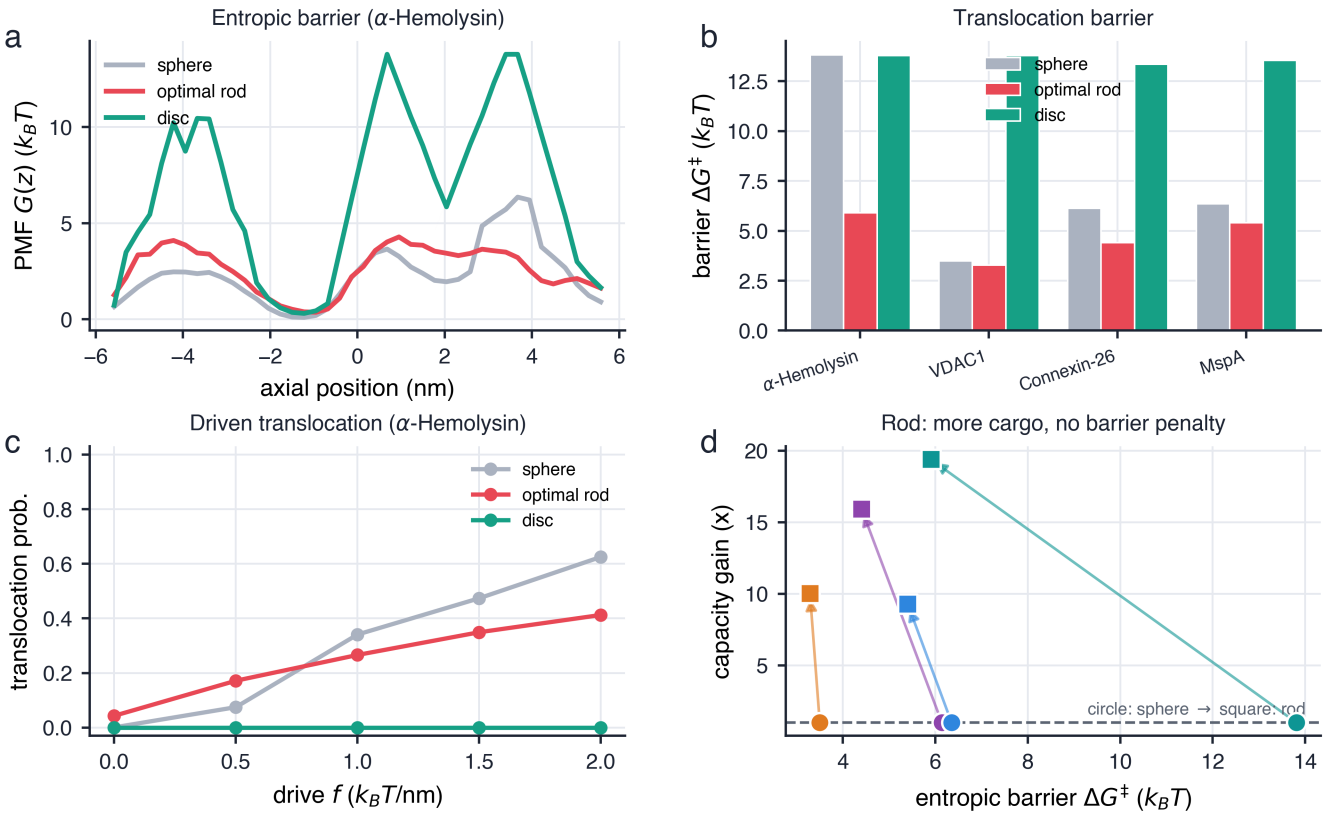


FIG. 8. **Brownian-dynamics translocation.** (a) Entropic PMF $G(z)$ for sphere, optimal rod and disc (α-hemolysin). (b) Peak translocation barrier per channel and shape; discs are blocked. (c) Driven translocation probability versus applied force. (d) Capacity gain versus translocation barrier: the sphere→rod transition (arrows) raises capacity by $\sim$10–19$\times$ with no barrier penalty.

fectively than spheres [1, 2, 5, 21]; our framework supplies a geometric, capacity-based account of why elongation helps, and quantifies how much. The disc/rod dichotomy we derive (discs for short, wide or slit-like barriers; rods for long channels) offers a concrete, structure-aware selection rule for the particle geometry best matched to a given biological gate.

Several limitations bound the scope of the present, deliberately minimal, model. (1) We treat particles and channels as rigid; deformability—central to the passage of cells and soft liposomes through constrictions—would relax the navigability bound and is a natural extension. (2) We resolve steric geometry only; electrostatics (the Debye layer in physiological salt), hydration shells and specific chemistry (e.g. the FG-nucleoporin mesh of the nuclear pore [22]) modulate real passage and would refine the effective aperture. (3) The transportable volume is a geometric capacity, and the Brownian-dynamics analysis an entropic, reduced-coordinate treatment; full three-dimensional rigid-body dynamics and free-energy methods would sharpen the kinetic predictions. (4) The mesoscale descriptors are order-of-magnitude, literature-informed estimates rather than structure-derived. None of these caveats affects the core geometric result, and each defines a concrete avenue for refinement. We emphasize that the framework is fully reproducible and structure-driven: every number derives from public coordinates through an open pipeline, and the central predictions are checked by independent collision detection.

The link to the moving-sofa problem is more than an analogy. The recent resolution of Moser's problem [11] establishes that the area-maximal navigating shape is anisotropic and corner-limited; our three-dimensional, volume-maximizing, structure-resolved formulation transfers that lesson to biophysics and turns it into a quantitative design tool. Beyond drug delivery, the same geometry governs the rational design of biological and solid-state nanopores for single-molecule sensing and sequencing [23, 24], where matching analyte shape to pore geometry is paramount.

X. METHODS

a. Structures. Coordinates were obtained from the Protein Data Bank (7AHL, 3EMN, 2ZW3; MspA from the 1UUN biological octamer). Heavy atoms were retained and assigned Bondi van der Waals radii [19] via

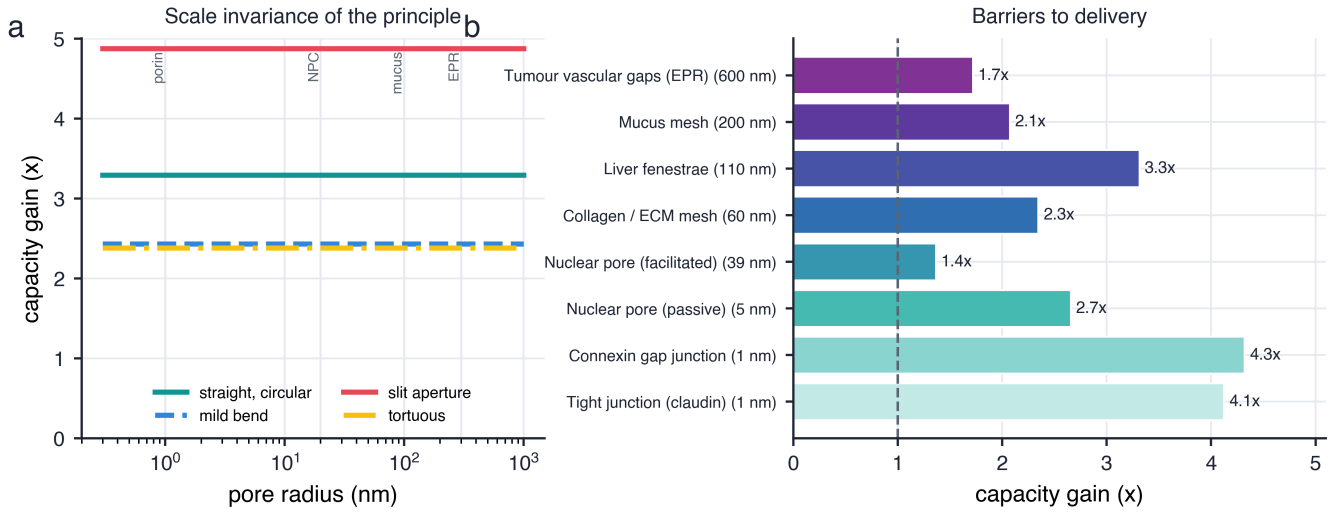


FIG. 9. **Mesoscale extension and scale invariance.** (a) For fixed shape ratios the capacity gain is independent of absolute pore radius across three decades—the principle is scale-invariant. (b) Predicted capacity gain for the major biological barriers to delivery; gains exceed unity throughout, with rods favoured in long channels and discs in short, wide windows.

Biopython [25].

b. Lumen geometry. The pore axis was initialized from a cylinder fit (minimizing the variance of atomic perpendicular distances) and refined to maximize the enclosed lumen length. The inscribed-sphere radius was propagated continuously along the axis from a central anchor using a bounded two-level grid search, with an eight-sector enclosure test halting propagation at the pore mouths, following the HOLE approach [12]. Tortuosity was computed from the smoothed centreline; aperture anisotropy from directional wall-clearance at the bottleneck.

c. Optimization. Equation (4) was solved by dense search over (ϵ_1, ϵ_2) and a transverse-scale/aspect parameterization, with the transverse-fit constraint enforced by boundary sampling and the navigability bound from Eq. (3). The volume law (2) was validated against 3×10^6-sample Monte-Carlo integration.

d. Atomistic validation. The optimal particle's surface was discretized and tested for overlap against all channel atoms (KD-tree nearest-atom queries) as it was (i) grown radially at the bottleneck (transverse limit) and (ii) elongated and swept along the lumen over a search of position and orientation (navigable length). The largest collision-free configuration defines the validated optimum.

e. Brownian dynamics. The entropic PMF $G(z) = -k_B T \ln \Omega(z)$ was estimated from the collision-free fraction $\Omega(z)$ of sampled rigid configurations [20]; overdamped Langevin trajectories were integrated on $G(z)$ with absorbing boundaries to obtain translocation probabilities and mean first-passage times. Figures were produced with Matplotlib; molecular surfaces use marching cubes [26]. SciPy [27] provided optimization and spatial routines. All code and derived data are available (see Data Availability).

XI. CONCLUSIONS

We have posed nanoparticle transport through a molecular channel as a three-dimensional moving-sofa problem and solved it on real, atomistically measured geometry. The optimal rigid carrier turns out to be a navigability-limited nanorod. It holds an order of magnitude more cargo than the largest sphere that fits, yet pays no penalty in translocation barrier. The same argument is scale-invariant by construction, so it applies equally to sub-nanometre porins and to micron-scale tissue gaps; the shape that wins simply shifts from a rod in long channels to a disc in short, wide windows. What began as a sixty-year-old geometry puzzle thus becomes a concrete design rule for nanocarriers and for the engineering of biological and solid-state nanopores.

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DATA AVAILABILITY

All atomic coordinates are public (PDB 7AHL, 3EMN, 2ZW3, 1UUN). The analysis pipeline, derived data and figure-generation code are available from the author on reasonable request.

Appendix A: Verification of the superquadric volume law

Equation (2) was checked against uniform Monte-Carlo integration over the bounding box for representative shapes. For the sphere ($\epsilon_1 = \epsilon_2 = 1$, $r_x = r_y = c = 1$) the formula gives 4.189 versus 4.189 (Monte-Carlo) and the exact $\frac{4}{3}\pi = 4.18879$; for a prolate ellipsoid, a unit cylinder, a flat disc and a short cylinder the formula and Monte-Carlo agree to better than 0.1%. An alternative single-exponent parameterization with the commonly mis-stated prefactor over-counts volume by a factor that grows with the exponent (e.g. 2.0× at $n = 2$, 6.8× at $n = 3$), which would spuriously favour boxy shapes; we therefore use the verified two-exponent law throughout.

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